

Dexter Williams

PHD STUDENT · INFORMATION

University of Wisconsin-Madison, Morgridge Hall, 1205 University Avenue, Madison, WI 53706

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Education

University of Wisconsin-Madison

PHD IN INFORMATION

- Advisor: Professor Jodi Schneider.
- Research Interests: Argument Mining, Computational Argumentation, AI and Law.
- 4.0 GPA

Madison, WI

Aug. 2025 - Present

University of Illinois Urbana-Champaign

PHD IN INFORMATICS

- Advisor: Professor Jodi Schneider.
- 4.0 GPA

Champaign, IL

Aug. 2024 - July 2025

University of St Andrews

BSC IN COMPUTER SCIENCE AND PHILOSOPHY (JOINT HONOURS)

- First Class (3.9 GPA).

St Andrews, Scotland, UK

Sep. 2018 - May 2022

Research Experience

Project Assistant: University of Wisconsin-Madison

SUPERVISOR: PROFESSOR JODI SCHNEIDER

- Support for a Second Edition of the Argumentation Mining monograph.
- Wrote an appendix for a book on argument mining, that documents existing corpora, and developed a website equivalent of the appendix.
- Acquired copyright permissions for the 100+ figures used in the book.
- Developed a data pipeline that measures the agreement of databases on the retraction status of scientific articles.

Madison, WI

2025-Present

Research Intern: Boise State University

SUPERVISOR: PROFESSOR CASEY KENNINGTON

- Joined the Retico team at Boise State University for 1 month to aid in the research and development of state-of-the-art incremental processing systems.
- Developed two new modules for Retico to handle speech diarization and automatic speech recognition.

Boise, ID

2025

Research Assistant: University of Illinois Urbana-Champaign

SUPERVISOR: PROFESSOR JODI SCHNEIDER

- Provided feedback for a new edition of a book on argument mining.
- Read and categorized 100+ papers on argument mining to assist in the writing of a new chapter.
- Funded by the University of Illinois at Urbana-Champaign Campus Research Board, \$14,714.

Champaign, IL

2025

Research Assistant: University of Illinois Urbana-Champaign

SUPERVISOR: PROFESSOR BERTRAM LUDÄSCHER

- Joined the explainable AI through computational argumentation (XAI-CA) project, which explores how argumentation can aid in explainability.
- Applied relation-based argument mining to US-legal documents to automatically construct structured argumentation frameworks.
- Funded by the 2025 Cardiff University-University of Illinois System Joint Research and Innovation Seed Grants Program.

Champaign, IL

2025

Undergraduate Research Assistant: University of St Andrews

SUPERVISOR: PROFESSOR IAN GENT

- Tested variants of Sudoku in a research project that aimed to provide human explanations of solutions to pen and paper puzzles.

St Andrews, Scotland, UK

2020-2021

Undergraduate Research Assistant: University of St Andrews

St Andrews, Scotland, UK

SUPERVISOR: DR. KASIM TERZIĆ

2019-2020

- Developed a program used by the Marine Biology department to automatically identify seal populations in aerial photographs.

Professional Experience

Software Engineer/Consultant

Bath, England, UK

CAPGEMINI ENGINEERING

2022-2024

- Joined the development team for a large rail project.
- Writing and testing the formal specification of the project.
- Implemented specification in Ada SPARK.
- Tested the project using a suite of tools developed under the HICLASS Research Programme.

Data Science and Innovation Intern

Leeds, England, UK

NATIONAL HEALTH SERVICE (NHS) DIGITAL

2021

- Joined the reproducible analytical pipelines team at NHS Digital for ten weeks.
- Produced four guides that teach good software engineering practices.
- Investigated existing PySpark libraries to find a simple and reproducible way for teams to implement basic metadata input validation in their projects.

Teaching Experience

Teaching Assistant: University of Illinois Urbana-Champaign

Champaign, IL

INFORMATION SCIENCE 203: ANALYTICAL FOUNDATIONS FOR INFORMATION PROBLEMS

2024-2025

- Two semesters teaching 80+ students introductory formal logic, discrete mathematics, and statistics.

Publications

CONFERENCE PAPERS

Williams, D., Liu, S., Stede, M., Wachsmuth, H., & Schneider, J. (2026). AMResources: Cataloging argument mining datasets. In M. Elaraby, A. Hautli-Janisz, J. Romberg, E. Musi, F. Ruggeri, & J. Lawrence (Eds.), Proceedings of the 13th Workshop on Argument Mining and Reasoning (pp. 52–58). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2026.argmining-1.5>

Williams, D. (2026). That feels convincing: Assessing the quality of pathotic arguments. Argumentation in the Digital Society. Proceedings of the 5th European Conference on Argumentation (ECA 2025). Forthcoming.

Williams, D. (2026). Decorum in the forum: Emotional interactions in U.S. presidential press conferences. Arguing Democracy. Proceedings of the 2025 NCA/AFA Argumentation Conference (Alta). Forthcoming.

Zheng, H., **Williams, D.**, & Ludascher, B. (2025). Using LLMs to model arguments in U.S. Supreme Court briefs: Preliminary report. Proceedings of the International Workshop on Translating Natural Legal Language into Formal Representation (NLL2FR 2025).

Zheng, H., **Williams, D.**, & Ludäscher, B. (2025). Modeling U.S. Supreme Court briefs with computational argumentation. In R. Markovich, L. D. Caro, A. Rapp, & C. Schifanella (Eds.), Legal Knowledge and Information Systems. SAGE Publications. <https://doi.org/10.3233/FAIA251631>

SELECTED NON-REFEREED/LIGHTLY REFEREED PUBLICATIONS, PRESENTATIONS & POSTERS

Williams, D., & Schneider, J. (2026). Emotional Argumentation in Second Look. Talk at The 2026 Symposium of the Applied Rhetoric Collaborative.

Williams, D., & Hatt, A. (2026). Don't Argue With Cops? . Poster at the University of Wisconsin-Madison May Day Tea 2026.

Williams, D. (2026). Computational Assessment of Pathotic Argumentation Quality. Online Handbook of Argumentation for AI, 5.

Williams, D. (2024). Assessing the Quality of Pathotic Arguments. Poster at the 2024 iSchool Research Showcase at the University of Illinois Urbana-Champaign.

Fellowships, Awards, & Honors

FELLOWSHIPS

2024	Informatics Recruitment Fellowship , University of Illinois Urbana-Champaign	\$ 30,000
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AWARDS

2025	iSchool Conference Presentation Award , University of Wisconsin-Madison	\$ 1,612
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2025	Informatics Conference Presentation Award , University of Illinois Urbana-Champaign	\$ 1,000
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HONORS

2018-2022	Dean's List , University of St Andrews
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Mentoring

2025-2026	Undergraduate Independent Study , Supervisor, University of Wisconsin-Madison
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2021	St Andrews Mentoring Initiative , Mentor, University of St Andrews
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2020	Coronavirus Tutoring Initiative , Tutor
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Service

PEER REVIEWER

11th International Conference on Computational Models of Argument (COMMA 2026)

15th edition of the Language Resources and Evaluation Conference (LREC 2026)

13th Workshop on Argument Mining (ArgMining 2026)

12th Workshop on Argument Mining (ArgMining 2025)